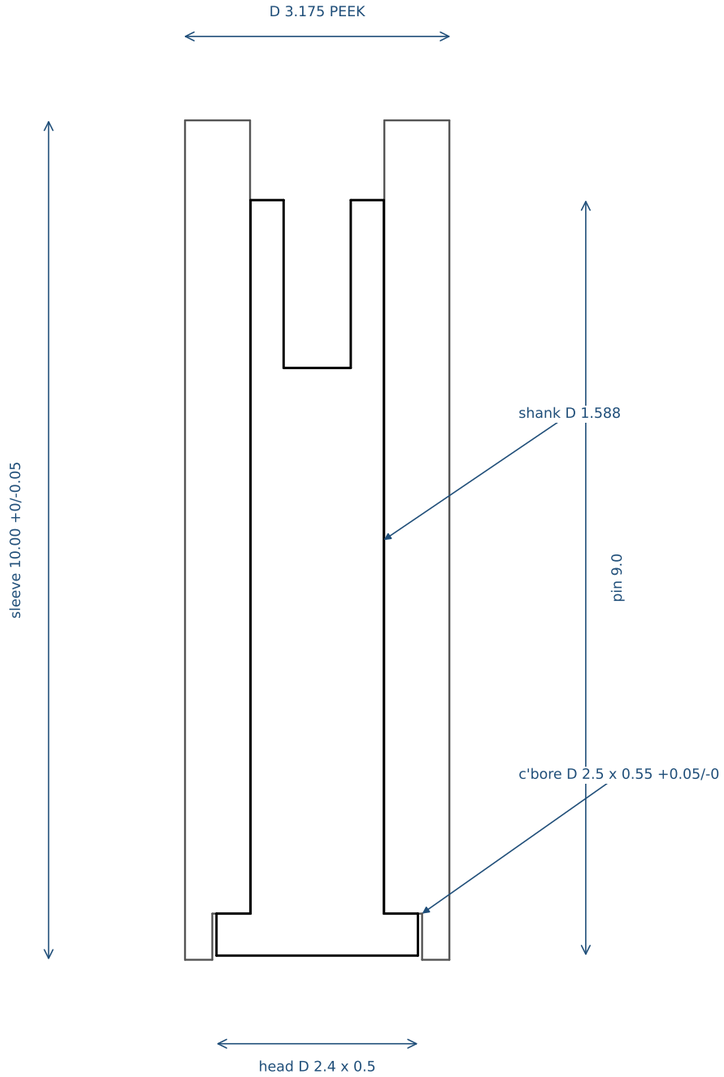
SECTION (pin black, sleeve grey), z = cap frame; stud tip at the bottom



NOTES:

1. POSITIVE STOPS (rev B): the contact spring pushes the pin INTO the cap, for ever. The sleeve seats on the square shoulder of the stud bore (D 3.15 to D 2.2); the pin HEAD seats on the counterbore floor in the sleeve. Neither can move inward. No Loctite 609, no staking (rev A; anaerobics do not bond PEEK).
2. Pin head face flat (or R 3 crown), square within 0.05, edge broken 0.05. Assembled, the gold face sits 0.05 +0.05/-0 BELOW the stud tip face.
3. Solder cup D 0.8 x 2.0 deep in the top end for the 30 AWG PTFE lead (pre-tin).
4. Assembly: solder the lead to the pin; feed the lead up through the sleeve, seat the head in the counterbore; feed the lead up the stud bore from the tip; press the sleeve home on the shoulder (0.025 interference in D 3.15). Sleeve end flush with the stud tip +0/-0.05.
5. Test: push 5 N on the pin face, no movement. Pull 2 N on the lead, pin stays seated. Pin to stud > 100 Mohm at 50 V DC. No solder or flux on the stud tip face.

RECORD SCONCE | CENTRE CONTACT PIN (headed) + PEEK SLEEVE (coax through the stud)

PART NO: RS-CAP-12L-PIN REV: B DATE: 2026-11-11 SUPERSEDES: rev A, A.1 (do not build or quote)

MATERIAL: PIN: brass C360, turned: head D 2.4 x 0.5, shank D 1.588 (1/16 in), 9.0 overall.

SLEEVE: natural PEEK, turned: OD 3.175 +0.01/-0, bore D 1.6 +0.03/-0, length 10.00 +0/-0.05, counterbore D 2.5 x 0.55 +0.05/-0 in the tip end.

FINISH: PIN: 1.3 um (50 uin) hard gold over 2.5 um nickel, all over (contact face + solder cup). SLEEVE: no finish.

UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown (enlarged)

GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted